Experiential Influences on the Development of Huddling Preferences and "Sibling" Recognition in Spiny Mice

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The role of experience in the development of huddling preferences and sibling recognition by spiny mouse (*Acomys cahirinus*) weanlings was investigated. In the 1st experiment, pups separated from their littermates and fostered onto unfamiliar females (and their offspring) on the day of birth, Day 10, or Day 20, subsequently showed preferential huddling with their foster littermates. Huddling by biological siblings was as infrequent as huddling by unrelated, unfamiliar agemates. The 2nd study revealed that the mother may play a role in the ontogeny of sibling recognition, possibly through distinctive labeling of her offspring. We conclude that recognition of littermates develops through experience in the home cage in the absence of an inborn ability to recognize one's biological siblings.

Excluding monozygotic multiple births (e.g., identical twins), the closest genetic relatives in mammalian species are full siblings as well as parents and offspring, for all of whom the coefficient of relatedness is .50 (Hamilton, 1964). By virtue of their genetic affinity, siblings afford unique opportunities for testing hypotheses stemming from theories of genetic influences on social behavior (e.g., kin selection and inclusive fitness theories). Furthermore, because littermate interactions may begin within minutes of birth, they should be of special interest for the study of the ontogeny of social behavior. We were surprised, therefore, that although parent-infant (especially mother-infant) interaction has been a favorite topic of animal behavior research for several decades, we found so few systematic investigations of interactions among siblings, regardless of species (see Bekoff, in press).

Recent observations of laboratory-reared spiny mice (*Acomys cahirinus*) have revealed that weanlings huddle preferentially with their (full-sibling) littermates, thereby indicating that littermates can be differentiated from unfamiliar agemates (Porter, Wyrick, & Pankey, 1978). Such selective huddling by littermates was subsequently found to be modifiable through post-weaning experience, however (Porter & Wyrick, 1979). Following an 8-day isolation period, or 4 days of exposure to

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unrelated agemates in the absence of the subjects' own littermates, sibling recognition was no longer evident. In the present series of studies, we investigated the influence of experience prior to weaning on the development of littermate recognition and huddling preferences. In particular, we assessed the possible roles of early experience and inborn determinants in huddling preferences along with the influence of the mother as a mediator of such preferences.

Experiment I: Biological Littermate vs Cagemate Preferences

In previous studies of sibling recognition in *A. cahirinus*, littermates remained together in their home cage, along with both parents, prior to weaning. Thus, subsequent preferences for littermates over unfamiliar agemates could have been the result of early experience in the homecage, an inborn ability to discriminate between siblings vs nonsiblings, or a combination of these 2 factors. Experiment I was designed to elucidate further the processes mediating the development of sibling recognition in *A. cahirinus*.

Method

Subjects

The spiny mouse (*Acomys cahirinus*) is a murid rodent that is indigenous to the Near East and gives birth to uniquely precocial offspring. An underlying assumption of our current research program is that *A. cahirinus* pups, because of their precocity, may be especially sensitive to early experimental manipulations. Although detailed studies of the natural history of *A. cahirinus* have yet to be conducted, this animal has been used in the authors' laboratory for several years in investigations of the development of social behavior and chemical communication. All subjects for the present series of experiments were born in the laboratory breeding colony where mated pairs were housed together continuously in individual breeding cages.

Apparatus

Animals for each of the experimental conditions were observed while housed in a 40-liter glass terrarium. A metal partition divided the terrarium into two 30 × 30-cm areas with a 7.6 × 7.6-cm-high opening centered at the bottom of the partition, allowing free movement between the 2 sides. The floor of this observation cage was covered with a layer of bedding material (sugar-cane waste) and Purina Mouse Chow was available ad libitum. Moisture was provided by fresh apple pieces that were placed in the cage at the end of the daily observation sessions.

Procedure

Fostering. At the appropriate age, 1 pup from each of 2 litters born on the same day was removed from its homecage and placed into the opposite cage containing an unfamiliar foster mother, adult male, and foster siblings. That is, single pups of the same age were exchanged between 2 litters, with the family units in each homecage otherwise remaining intact. These pups remained with their foster families with no further manipulations until testing. All litters contained from 2 to 4 pups. Fostered pups were identified by ear punches.

Animals were randomly assigned to 3 conditions that varied according to the age at which pups were cross-fostered. In the Day-of-Birth (DOB) condition, 10 pairs of

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pups (from 20 different litters) were cross-fostered in the above manner within 12 hr of birth. Likewise, 10 pairs of pups were cross-fostered on postnatal Day 10 of life (with the day of birth being Day 1), and 10 pairs were cross-fostered on Day 20. In each instance, all pups had remained in their respective home cages with their biological parents and littermates until being fostered.

Testing. Testing began on postnatal Days 28-34 with all 4 animals in a given observation cage being the same age. The mean ages at the start of testing for the 3 conditions were: DOB condition, 30.5 days; Day 10 condition, 30.7 days; Day 20 condition, 30.4 days.

On the 1st day of testing, the 4 weanlings assigned to a given observation group were removed from their respective cages, marked for purposes of identification, and placed together into an observation terrarium where they remained throughout the 5-day observation period. Weanlings in each group were observed during six 5-min sessions on each of the 5 days. The 1st observation session (on Day 1) began 60 min after the animals had been placed in the observation cage to allow time for them to explore and become somewhat familiar with their new environment. At the beginning of each 5-min observation session and at the end of each 60-sec interval therein, the identities of all animals that were physically touching one another (touching of limbs, head, or trunk, excluding touching of whiskers or tails) were recorded. Thus, each 5-min observation session had a total of 6 observation points. The six 5-min observation sessions were distributed throughout the day with at least 30 min between successive sessions. Over the 5 days of observation each group had a total of $36 \times 5 = 180$ sample points.

Each observation group consisted of 2 pups that had been exchanged between 2 age-matched litters, along with 1 pup (foster sibling) from each of the 2 litters into which these pups had been fostered. Thus, a given group contained 2 pairs of biological siblings (littermates) which had been reared apart from the time of the cross-fostering manipulation. As an example, at birth Litters A and B consisted of pups A1, A2, and A3, and B1, B2, and B3, respectively. At the appropriate age, single pups from each litter were exchanged so that A1, A3, and B1 were subsequently reared together in 1 cage (along with both A parents), and pups B2, B3, and A2 were reared together in a 2nd cage. At the beginning of the test period, A1, A2, B1, and B2 were placed together in a single observation terrarium. In this testing situation, therefore, 3 classes of dyadic pairings were possible:

Biological siblings: A1, A2, and B1, B2.

Foster littermates (unrelated cagemates): A1, B1 and A2, B2.

Unrelated, unfamiliar weanlings: A1, B2 and A2, B1.

Ten groups of 4 animals each were observed in this manner in each of the 3 experimental conditions.

Results

As in our earlier studies of sibling recognition in *A. cahirinus*, relative frequency of dyadic pairing was adopted as an unambiguous index of social preferences. Recognition was operationally defined as a greater relative frequency of pairing involving specific animals than would be expected by chance alone.

The 1st few hours following introduction into the observation cage were typically the period of greatest activity during the 5 days of observation. At that time, brief encounters involving sniffing were frequent, with animals withdrawing if rapidly approached by another. By the 2nd day, however, animals showed little activity during the observation sessions; regardless of whether paired or not, they usually sat quietly

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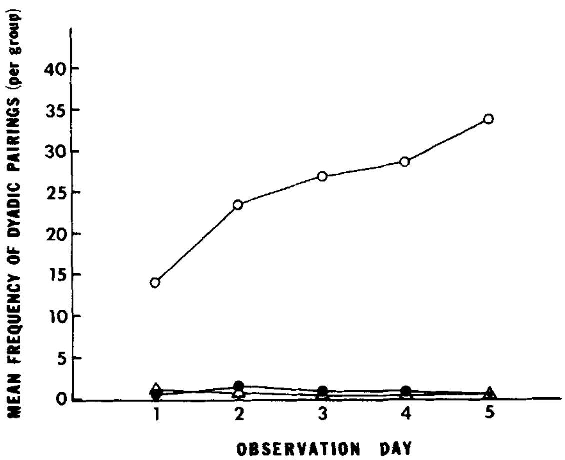

Fig. 1. Mean daily frequencies of dyadic pairing (per group) for the 3 fostering conditions. ○, Foster littermates; ●, Biological siblings; △, Unrelated, unfamiliar animals.

with little overt behavior being displayed. The frequency of foster littermate pairings showed a marked increment from Day 1 to 2 with a more gradual increase over the remainder of the observation period (Fig. 1). However, regardless of the age at fostering, pairings by unrelated foster littermates (pups housed together following fostering) were significantly more frequent than were dyadic pairings by biological siblings reared apart after fostering, or by unrelated unfamiliar weanlings (Table 1). Post hoc analysis revealed that frequencies of pairing by biological siblings vs unrelated unfamiliar animals did not differ statistically within any of the 3 fostering conditions.

Experiment II: Maternal Labeling and Littermate Recognition

In Experiment I, as in previous studies of sibling recognition in *A. cahirinus*, littermate exposure was not separated from exposure to the lactating female (biological or foster mother). Therefore, postweaning preferences for littermates could have been mediated to some extent by the mother who functioned as a common

TABLE 1. Mean Frequencies of Dyadic Pairing by Biological Siblings, Foster Littermates (i.e., Unrelated Cagemates), and Unrelated, Unfamiliar Weanlings for Each of the 3 Fostering Conditions.

	Composition of Dyads		
Condition (Day of Fostering)	Biological Siblings	Foster Littermates	Unrelated, Unfamiliar Weanlings
Day 1 (day of birth) (n = 10 groups)	4.9	173.5 $F = 61.78$, df 2/18, $p < .0001$	3.1
Day 10 (n = 10 groups)	2.3	115.3 $F = 15.33$, df 2/18, $p < .005$	2.2
Day 20 (N = 10 groups)	7.6	85.4 $F = 12.08$, df 2/18, $p < .001$	6.1

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source of distinctive stimuli for all of the pups suckling from her. The purpose of Experiment II was to test for the influence of the mother in littermate recognition and huddling preferences in *A. cahirinus*.

Method

Pups born of multiparous females were randomly assigned to 1 of 2 treatment conditions.

Siblings Reared Apart but Nursed by the Same Lactating Female

On Day 4 of life, 2 siblings from the same litter were removed from their home-cage and placed separately in individual cages measuring 28.5 × 18 × 12.5 cm containing ad lib food and water. The biological mother of these 2 pups was moved between the 2 cages housing her offspring at 12-hr intervals, remaining in the cage of 1 pup from 0800 to 2000 hours each day and from 2000 to 0800 hours in the cage containing the 2nd pup. The female was switched back and forth between her 2 offspring in this manner from Day 4 postpartum until testing. The 2 siblings in each pair were therefore reared apart from one another after Day 4 but were exposed to the same lactating female (the biological mother) at 12-hr intervals throughout the treatment period.

Testing was conducted over a 5-day period (Monday through Friday) beginning when the pups were 17-21 days of age (i.e., 13-17 days after the start of the experimental manipulation). At the start of testing, 2 pairs of littermates (a total of 4 pups born to 2 different females) treated in the above manner were placed together into a single observation terrarium. All 4 pups assigned to a given observation group had been born on the same day. The same observation procedure as described for Experiment I was then adopted to record instances of dyadic touching. Eleven groups consisting of 4 pups each were tested in Condition 1.

Siblings Reared Apart and Nursed by Different Lactating Females

A total of 44 pups was assigned to the 2nd condition (11 groups of 4 pups each). Each observation group was made up of 2 pairs of siblings that had been born on the same day. On postnatal Day 4, the 4 pups in a given observation group were removed from their respective home cages and placed into individual cages where they remained until the beginning of testing. Unlike Condition 1, however, littermates were not nursed by the same female during the treatment period. Rather, 1 pup in each sibling pair was exposed to its own (biological) mother for 12 hr each day (either 0800-2000 hours, or 2000-0800 hours) while its littermate was exposed to the mother of the other litter assigned to that same group. For example, litters born to Females A and B were each comprised of 2 offspring, al, a2 and b1, b2, respectively. Beginning on Day 4, Female A was switched between 1 of her own pups (al) and 1 pup from the other litter (b1) every 12 hr, while Female B was switched between pup a2 (other litter) and b2 (own). Thus, 2 unrelated pups that had never been directly exposed to one another were separately housed with the same lactating female 12 hr each day, while biological littermates were exposed to 2 different females.

As for Condition 1, all 4 pups assigned to a given group were placed into an observation cage on Days 17-21 of life for data collection over a period of 5 consecutive days. Further procedural details were otherwise identical to those of the previously described condition

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Results

Siblings Reared Apart but Nursed by the Same Lactating Female

Across the 11 groups of animals in Condition 1, the mean frequencies of observed dyadic pairings of siblings (nursed by the same lactating female) as compared to unfamiliar nonsiblings (nursed by different lactating females) were 63.1 and 11.6, respectively. Because nonsibling pairings could result from twice as many combinations of 2 animals (in each observation cage) as could sibling pairing, the expected ratio of nonsibling to sibling pairings was 2:1 within each group. To correct for this expected 2:1 ratio, the frequency of nonsibling pairings for each group was multiplied by a factor of .75 and the frequency of sibling pairings by 1.5. The corrected mean frequencies of nonsibling versus sibling pairings were then compared using the t-test for correlated samples. The results of this analysis indicate that pairings by biological siblings were significantly more frequent than were the pairings by the nonsiblings ($t = 2.09$, $df = 10$, 1-tailed $p < .05$).

Siblings Reared Apart and Nursed by Different Lactating Females

Within Condition 2, 3 equally probable (by chance alone) classes of dyadic pairing could occur. The mean frequencies of pairing (per group) over the 5-day observation period for each of these classes were as follows: biological siblings nursed by different females, $X = .64$; nonsiblings nursed by the same lactating female, $X = 18.9$; nonsiblings nursed by different lactating females, $X = .00$. These means indicate that dyadic pairings were relatively infrequent across the three conditions. Because of the prevalence of "0" scores, the Sign Test was used to compare the frequencies of pairing by biological siblings nursed by different females and pairings by nonsiblings nursed by the same female to those of nonsiblings nursed by different females. The *a priori* prediction was that pairings by the former 2 classes of pups should each be more frequent than pairings involving nonsiblings nursed by different females. The results of the Sign Tests confirmed 1 of the above predictions, that is, that the frequency of dyadic pairing by nonsiblings sharing the same mother was significantly greater than for nonsiblings reared by different females ($p = .03$, 1-tailed test). No difference was found between the frequencies of dyadic pairing by siblings nursed by different females and nonsiblings reared by different females ($p = .25$, 1-tailed test).

Between-Condition Comparisons

The mean frequency of dyadic pairing by biological siblings nursed by the same lactating female ($X = 63.1$; Condition 1) was significantly greater than that for siblings nursed by 2 different females ($X = .6$; Condition 2) or the frequency of pairings by nonsiblings nursed by the same female ($X = 18.9$; Condition 2). Results of respective Mann-Whitney U tests for these 2 sets of comparisons were as follows: $Z = 2.94$, $p < .005$, 2-tailed test; $Z = 1.67$, $p < .05$, 1-tailed test.

Discussion

The data resulting from the above experiments support the conclusion that littermate recognition in *A. cahirinus* develops through experience, while offering no evidence to support inborn recognition of biological siblings. This is in marked contrast to a recent report of incubator-reared pigtail macaques showing preferences for paternal half-siblings with whom they had no previous experience (Wu, Holmes, Medina, & Sackett, 1980).

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In Experiment I, pups that were fostered on the Day of Birth, Day 10, or Day 20 of life subsequently huddled preferentially with their foster littermates rather than with their biological siblings during testing that began on Days 28-34. Furthermore, biological siblings were not preferred over unrelated, unfamiliar agemates, despite the siblings having been raised together for up to 20 days prior to fostering. This latter finding corroborates and extends that of previous research (Porter & Wyrick, 1979) showing that in an undisturbed situation, littermates develop preferences for one another by 14-16 days of age. Complete isolation for 8 days postweaning, however, results in a loss of such sibling preferences (i.e., sibling huddling being no more frequent than huddling by unfamiliar pups).

A somewhat similar experientially mediated reversal of early preferences was found by Doane and Porter (1978) in their study of dietary influences on the attractiveness of maternal chemical cues in this same species. Athough 1-day-old spiny mouse pups typically respond preferentially to chemical cues produced by lactating females fed the same diet as their mother, they will gradually develop a preference for chemical cues associated with the diet of their foster mother (following fostering on the 2nd day of life) over those of their biological mother's diet. Therefore although recognition of littermates, like preferences for specific maternal chemical cues, may develop from early experience in the homecage, such effects are not immutable and can be modified or reversed through subsequent environmental events.

Detailed studies of rat pups have likewise implicated the role of experiential factors in the development of huddling preferences. At 15-20 days of age, rats have been found to huddle preferentially with anesthetized (unfamiliar) prepubertal conspecifics scented with odorants to which the subjects had been exposed in their nest (Brunjes & Alberts, 1979). A systematic comparison of the ontogeny of huddling by (altricial) rats and (percocial) spiny mice would be of considerable interest in the light of the differences in sensory and behavioral development of the neonates of these 2 species.

For mammalian neonates, the most important feature of their environment is without doubt the mother, which provides nurturance and protection. As seen in Experiment II, the mother may also have an influence on the development of sibling recognition. Biological siblings that are separated on Day 4 of life but continue to share the same mother subsequently show evidence of recognition and discriminative interactions. If nursed by different females following separation, siblings do not recognize one another, thereby demonstrating that relatively brief early sibling exposure in the homecage is not itself sufficient for subsequent recognition. Sharing a mother in common, however, does appear to be sufficient for the development of pup recognition. When nonsiblings with no prior exposure to one another shared the same mother (lactating female), they subsequently displayed significantly more dyadic pairings than did unfamiliar nonsiblings nursing from different females. Such exposure to a common mother does not result in the same degree of preferential huddling as does common maternal exposure coupled with early exposure of the subject pups to one another, however (see between-condition comparisons of Experiment II).

In the litter situation, therefore, maternal influences may interact with early exposure of the pups (to one another) to mediate subsequent littermate recognition. Although the actual process through which pups acquire distinctive cues associated with their mother is not known, it may be similar to "maternal labeling" in goats whereby mothers are believed to transfer odor cues directly to their offspring by licking the body surface, or indirectly through their milk (Gubernick, 1980).

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The lack of evidence for an inborn ability to recognize one's siblings should not be interpreted as arguing against the Theory of Kin Selection. Although the recognition of cues associated with one's littermates appears to be acquired through experience, the tendency to respond preferentially to individuals possessing such familiar characteristics may itself reflect an inborn predisposition. Also, if littermates typically remain together and share the same mother(s) prior to weaning, then sibling recognition could develop through experience without the need for genetic coding of sibling characteristics (see Bekoff, in press). In such a system, discriminative interactions with individual agemates with whom a pup was raised would constitute preferential interactions between siblings and could therefore evolve through kin selection.

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